DDA3020 Machine Learning
Lecture 02 Probability Theory

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Outline

1. A brief review of last week
2. Probability, event, random variables
3. Probability of discrete random variables
4. Probability of continuous random variables
5. Some common discrete distributions
6. Some common continuous distributions
7. Information theory

Definition and branches of machine learning

Basic concepts, learning process

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Random experiment, sample space, event

- Random experiment: we describe a random experiment by its **procedure** and observations of its **outcomes**. For example, we toss a coin 2 times, and observe which side is up after each toss.
- Sample space: All possible outcomes of the random experiment form a sample space S. For the above coin toss example, we define

$$S = \{(Head, Head), (Head, Tail), (Tail, Head), (Tail, Tail)\}.$$

- Event: A **subset** of sample space S, denoted as A, can be called as an event in a random experiment, *i.e.*, $A \subset S$. In the above example, we define an event A as *at least one head up*, then it can be represented by

$$A = \{(Head, Head), (Head, Tail), (Tail, Head)\} \subset S.$$

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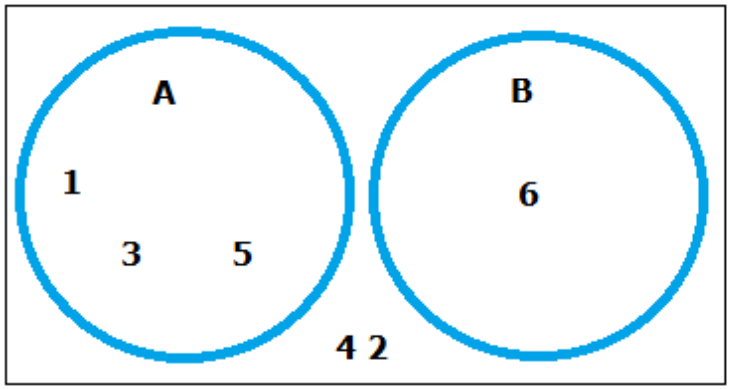
Probability of events

Assuming events $A \subset S$ and $B \subset S$, the probabilities of events related with and must satisfy,

- $P(A) \geq 0$
- $P(S) = 1$
- If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$;
  otherwise, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

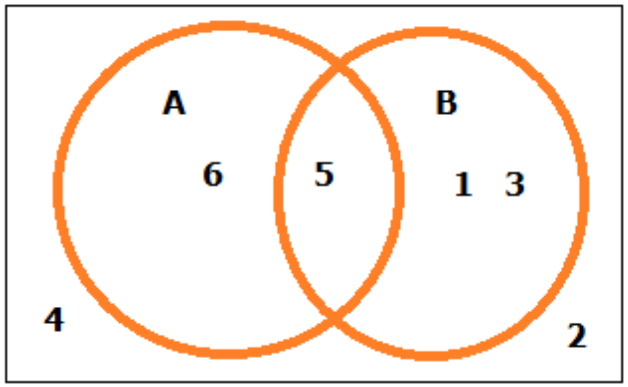
Disjoint Events

Event A: Get an odd Number
Event B: Get a 6



Overlapping Events

Event A: Get a number over 4
Event B: Get an odd number



Random variables

- A random variable is a real valued function from the sample space S to a real space $\mathbb{R}$, as follows:

$$X : S \to \mathbb{R}$$

- Still take the 2-times coin toss as example, if we define the random variable as the number of tails, then we have

$$X((H,H)) = 0, X((H,T)) = 1, X((T,H)) = 1, X((T,T)) = 2.$$

  Then, the output space of X is denoted as $\{0,1,2\}$, also called state space $\mathcal{X}$.
- There are two types of random variables:
  - **Discrete**: $\mathcal{X}$ is discrete
  - **Continuous**: $\mathcal{X}$ is continuous

Exercises of Random variables

- Exercise 1: For the random experiment of throwing a pair of dice, please write down its sample space, and the event *at least one number 3 on two dice* as well as its probability. $\{3, x\}$ $\{y, 3\}$ $\#3$ $\frac{11}{36}$

- Exercise 2: For the above random experiment, we define the random variable as *the number summation of two dice*, please write down its state space. $\{2, 3 \ldots 12\}$

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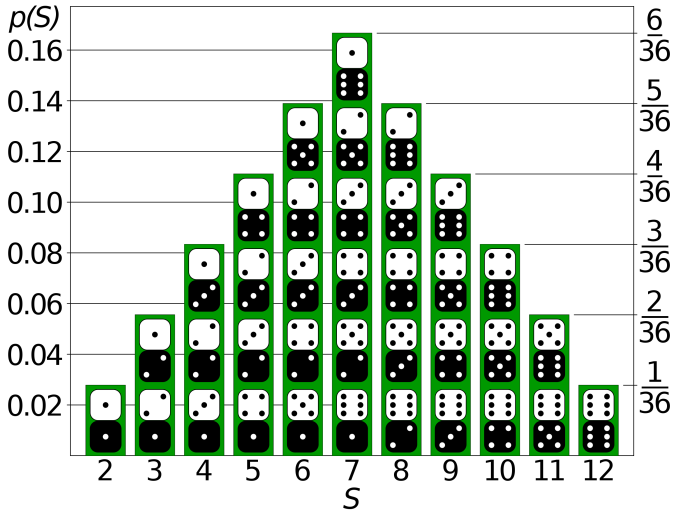
7 Information theory

Probability of discrete random variables

- Probability of discrete random variable describes the chance of each state x in $\mathcal{X}$ for random variable X in a random experiment, denoted as

$$P(X = x), x \in \mathcal{X}.$$

- Exercise 1: If we assume the coin is fair in the random experiment of 2-times coin toss, *i.e.*, $P(Head) = P(Tail) = \frac{1}{2}$ for toss, please compute the probability of different number of tails.
- Exercise 2: For the random experiment of throwing a pair of dice, please compute the probability of the state of 3 (*i.e.*, the number summation of two dice equals to 3).



Joint, marginal, conditional probability

- Probability of a union of two events: Given two events A and B, we define the probability of A or B as follows:

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B), \\ &= P(A) + P(B) \text{ if } A \text{ and } B \text{ are } \textbf{mutually exclusive}. \end{aligned} \tag{1}$$

- Joint probabilities: The probability of the joint event A and B is defined as follows:

$$P(A, B) = P(A \cap B) = P(A|B)P(B) = P(B|A)P(A), \tag{2}$$

  It is called the **product rule**.
- Marginal distribution: Given the above joint distribution, we can define the **marginal distribution** as follows:

$$P(A) = \sum_b P(A, B) = \sum_b P(A|B = b)P(B = b), \tag{3}$$

  which sums over all possible states of B. It is called the **sum rule**.

Conditional probability and Bayes rule

- Conditional probability: Recalculating probability of event A after someone tells you that event B happened, as follows:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{4}$$

$$P(A \cap B) = P(A|B)P(B) \tag{5}$$

- Bayes Rule: Combining the definition of conditional probability with the product and sum rules yields Bayes rule, as follows:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}, \tag{6}$$

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P(X = x)P(Y = y|X = x)}{\sum_{x' \in \mathcal{X}} P(X = x')P(Y = y|X = x')} \tag{7}$$

Application of Bayes rule: medical diagnosis

- Suppose that you do a medical test for breast cancer, the test result could be *positive* or *negative*. We denote $x = 1$ as the event of positive test, while $x = 0$ as the event of negative test. We denote $y = 1$ as the event of having breast cancer, while $y = 0$ as the event of no breast cancer.
- Suppose that if one has breast cancer, the test will be positive with the probability 0.8, *i.e.*,

$$P(x = 1|y = 1) = 0.8. \tag{8}$$

- Then, if one gets a positive test result, what is the probability of having breast cancer? $P(y = 1|x = 1) = 0.8$?

Application of Bayes rule: medical diagnosis

- It is WRONG! It ignores the prior probability of having breast cancer.
- According to statistics, the average risk of a woman in the United States developing breast cancer sometime in her life is about 13%, *i.e.*,

$$P(y = 1) = 0.13. \tag{9}$$

- We also need to take into account the fact that the test may be a **false positive** or **false alarm**. Unfortunately, such false positives are quite likely (with current screening technology):

$$P(x = 1|y = 0) = 0.1. \tag{10}$$

- Combining all above probabilities using Bayes rule, we can compute

$$\begin{aligned} P(y = 1|x = 1) &= \frac{P(x = 1|y = 1)P(y = 1)}{P(x = 1|y = 1)P(y = 1) + P(x = 1|y = 0)P(y = 0)} \\ &= \frac{0.8 \times 0.13}{0.8 \times 0.13 + 0.1 \times 0.87} = 0.5445. \end{aligned} \tag{11}$$

  It tells that if you test positive, you have have about a 54% chance of really having breast cancer!

Independent random variables

- Independent: If X and Y are independent, denoted as $X \perp Y$, then the joint probability can be represented as the product of two marginals, *i.e.*,

$$X \perp Y \iff P(X, Y) = P(X)P(Y). \tag{12}$$

- Given the above independence, we can use fewer parameters to define a joint probability. Suppose that X has 3 states, Y has 4 states, then we need $3 - 1 = 2$ and $4 - 1 = 3$ free parameters to define $P(X)$ and $P(Y)$, respectively.
- If without the independence, how many free parameters do we need to define the joint probability $P(X, Y)$? $(3 \times 4) - 1 = 11$.
- If given the independence, *i.e.*, $P(X, Y) = P(X)P(Y)$, how many free parameters do we need? $(3 - 1) + (4 - 1) = 5$.

Expectation and variance of discrete random variables

- Expectation (or mean): $E(X) = \sum_{x \in \mathcal{X}} xP(X = x)$
- Expectation of a function: $E(f(X)) = \sum_{x \in \mathcal{X}} f(x)P(X = x)$
- Moments: expectation of power of X: $M_k = E(X^k)$
- Variance: Average (squared) fluctuation from the mean

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2 = M_2 - M_1^2. \tag{13}$$

- Standard deviation: Square root of variance, *i.e.*,

$$\text{Std} = \sqrt{\text{Var}(X)}. \tag{14}$$

- `Exercise` : For the random variable of the number of tails in the random experiment of 2-times coin toss, please compute the above values.

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Continuous random variables

- A random variable X is continuous if its state space $\mathcal{X}$ is uncountable.
- In this case, $P(X = x) = 0$ for each x.
- If $p_X(x)$ is a probability density function (PDF) for X, then

$$P(a < X < b) = \int_a^b p(x)dx \tag{15}$$

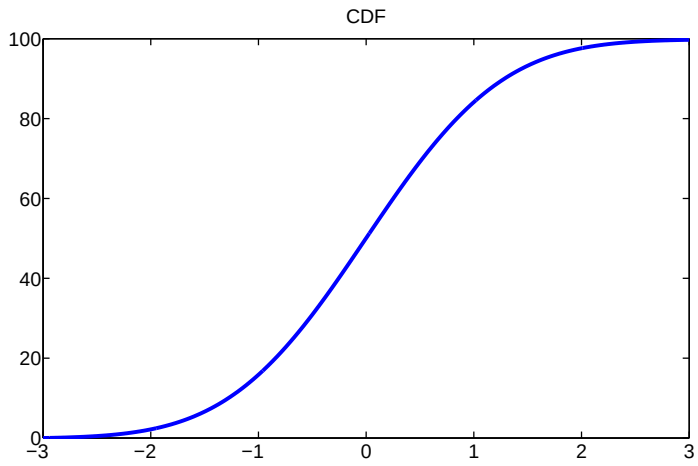
$$P(a < X < a + dx) \approx p(a) \cdot dx \tag{16}$$

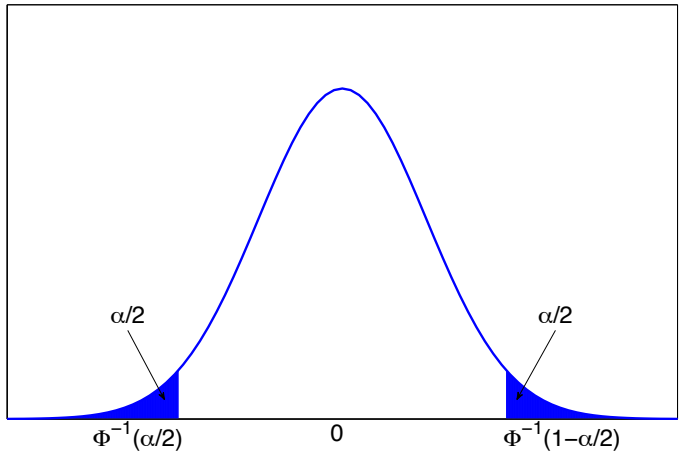
- The cumulative distribution function (CDF) is $F_X(x) = P(X < x)$. We have that $p_X(x) = F'(x)$, and $F(x) = \int_{-\infty}^{x} p(s)ds$.

Bivariate continuous distributions: Marginalization, Conditioning and Independence

- $p_{X,Y}(x, y)$, joint probablity density function of X and Y
- $\int_x \int_y p(x, y) dx dy = 1$
- Marginal distribution: $p(x) = \int_{-\infty}^{\infty} p(x, y) dy$
- Conditional distribution: $p(x|y) = \frac{p(x,y)}{p(y)}$
- Note: $P(Y = y) = 0$! Formally, conditional probability in the continuous case can be derived using infinitesimal events.
- Independence: X and Y are independent if $p_{X,Y}(x, y) = p_X(x) p_Y(y)$

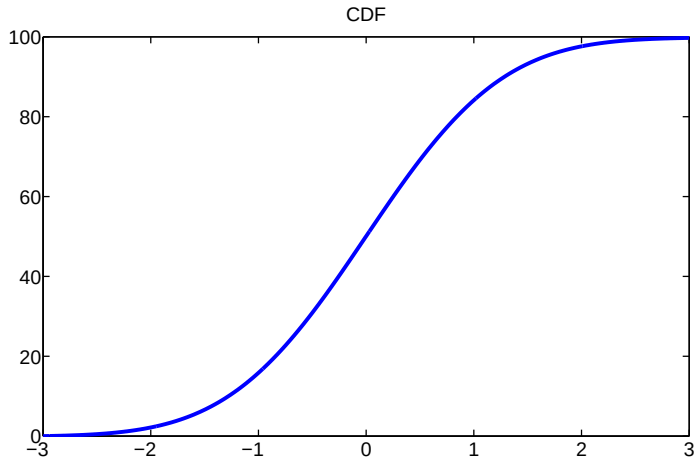
Quantiles

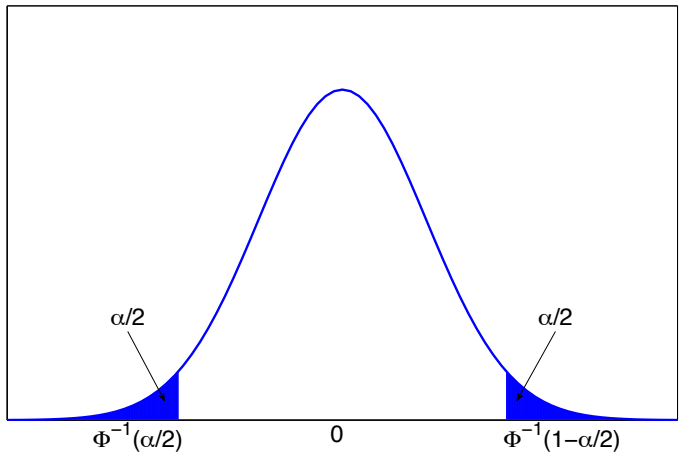




- Since the CDF $F(\cdot)$ is a monotonically increasing function, it has an inverse; let us denote this by $F^{-1}(\cdot)$.
- If $F(x)$ is the CDF of X, then $F^{-1}(\alpha)$ is the value of x_α such that $P(X \leq x_\alpha) = \alpha$; this is called the a quantile of F. The value $F^{-1}(0.5)$ is the median of the distribution, with half of the probability mass on the left, and half on the right. The values $F^{-1}(0.25)$ and $F^{-1}(0.75)$ are the **lower** and **upper quartiles**.

Quantiles





- We can also use the inverse CDF to compute tail area probabilities.
- For example, if Φ is the CDF of the Gaussian distribution $\mathcal{N}(0,1)$, then points to the left of $\Phi^{-1}(\alpha/2)$ contain $\alpha/2$ probability mass. By symmetry, points to the right of $\Phi^{-1}(1-\alpha/2)$ also contain $\alpha/2$ probability mass.
- Hence, the central interval $(\Phi^{-1}(\alpha/2), \Phi^{-1}(1-\alpha/2))$ contains $1-\alpha$ of the mass. If we set $\alpha = 0.05$, the central 95% interval is covered by the range

$$(\Phi^{-1}(0.025), \Phi^{-1}(0.975)) = (-1.96, 1.96). \tag{17}$$

For a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$, the central 95% interval is $(\mu - 1.96\sigma, \mu + 1.96\sigma)$.

Expectation and variance of continuous random variables

Similar to that of discrete random variables, only change the summation $\sum$ to the integral $\int$.

- Expectation (or mean): $\mu = E(X) = \int_{\mathcal{X}} x \cdot p(x)dx$
- Moments: expectation of power of X: $M_k = E(X^k) = \int_{\mathcal{X}} x^k \cdot p(x)dx$
- Variance: Average (squared) fluctuation from the mean

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - E(X)^2 = M_2 - M_1^2. \tag{18}$$

- Standard deviation: Square root of variance, *i.e.*,

$$\text{Std} = \sqrt{\text{Var}(X)}. \tag{19}$$

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Binary variables (discrete r.v.)

- We firstly consider the probability of a binary random variable $x \in \{0, 1\}$. Suppose that you toss a coin, and $x = 1$ denotes the event of 'heads', while $x = 0$ indicates the event of 'tails'.
- The probability of $x = 1$ is described by a parameter μ,

$$p(x = 1|\mu) = \mu, \tag{20}$$

  where $\mu \in [0, 1]$, and we can obtain that $p(x = 0|\mu) = 1 - \mu$.
- The probability distribution over x can therefore be written in the form

$$\text{Bern}(x|\mu) = \mu^x(1 - \mu)^{1-x}, \tag{21}$$

  which is called Bernoulli distribution.
- Its mean and variance are

$$\mathbb{E}[x] = \sum_x x\text{Bern}(x|\mu) = \mu, \tag{22}$$

$$\text{var}[x] = \mathbb{E}[(x - \mu)^2] = \mu(1 - \mu) \tag{23}$$

Binomial variables

- Imagine that you toss the coin N times, and each tossing follows the Bernoulli distribution $p(x|\mu)$. We denote the variable m as the numbers of heads, then its distribution is formulated as follows:

$$\text{Bin}(m|N,\mu) = \binom{N}{m}\mu^m(1-\mu)^{N-m}, \tag{24}$$

  which is called Binomial distribution, where

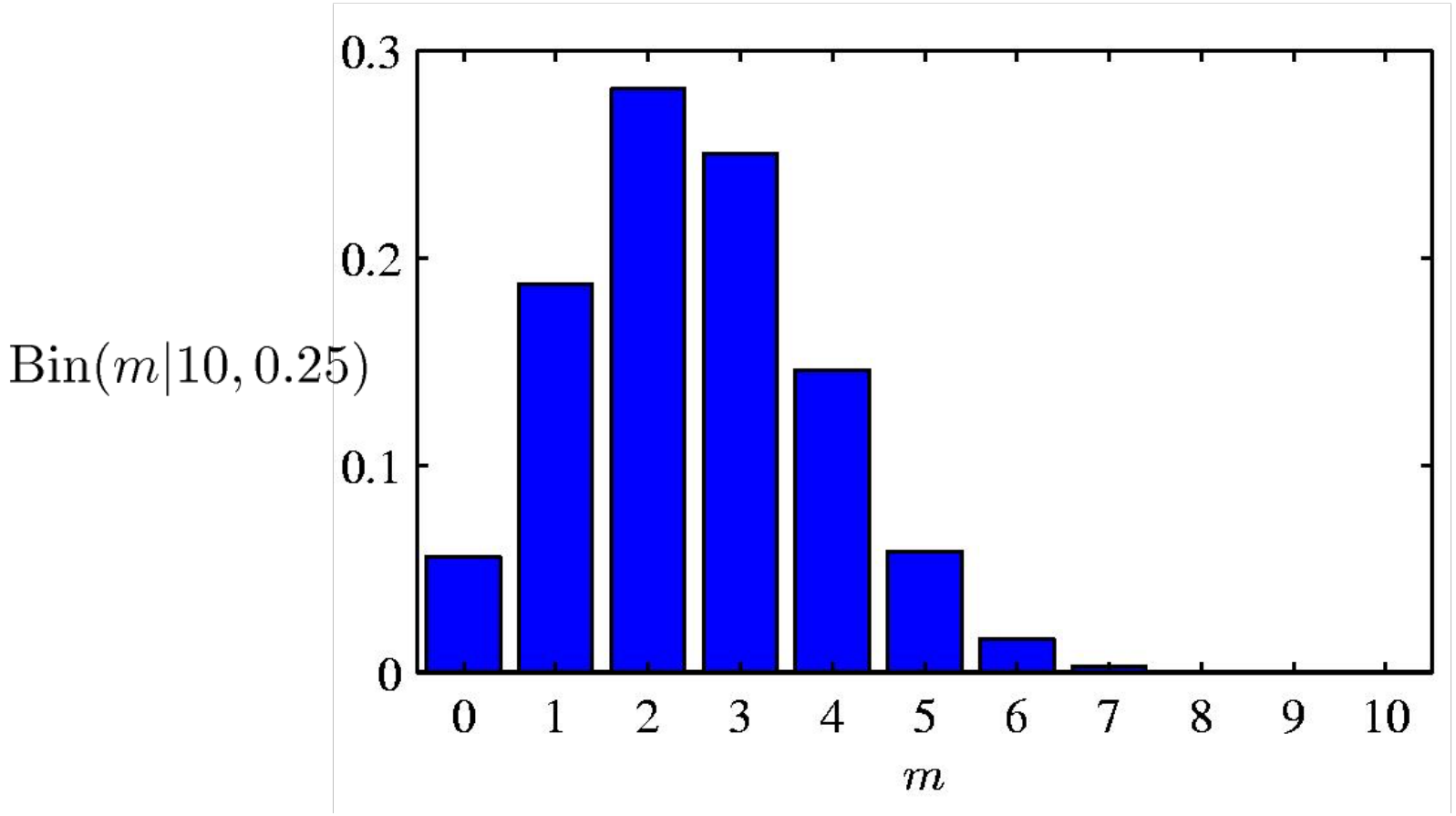
$$\binom{N}{m} = \frac{N!}{(N-m)!m!}. \tag{25}$$

- Its mean and variance are

$$\mathbb{E}[m] = \sum_{m=0}^{N} m\text{Bin}(m|N,\mu) = N\mu, \tag{26}$$

$$\text{var}[m] = \mathbb{E}[(m-N\mu)^2] = N\mu(1-\mu). \tag{27}$$

Binomial distribution



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Gaussian distribution (continuous r.v.)

- The Gaussian, also known as the **normal** distribution, is a widely used model for the distribution of **continuous** variables. In the case of a single variable x, the Gaussian distribution can be written in the form

$$\mathcal{N}(x|\mu,\sigma) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{28}$$

  where μ is the **mean** and σ^2 is the **variance**.

- For a D-dimensional vector $\boldsymbol{x}$, the multivariate Gaussian distribution takes the form

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}{2}\right), \tag{29}$$

  where $\boldsymbol{\mu}$ is a D-dimensional **mean vector**, and $\boldsymbol{\Sigma}$ is a $D \times D$ **covariance matrix**, and $|\boldsymbol{\Sigma}|$ denotes the determinant of $\boldsymbol{\Sigma}$.

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What is information

- To know what is information, we need to know a bit who is Claude Shannon and his information theory (let's see a video).
- Shannon defined the measure that quantifies the uncertainty of an event with given probability - a bit.
- For a discrete random variable (a source) with finite alphabet, as follows

$$\mathcal{X} = \{x_0, \cdots, x_k, \cdots x_S\},$$

where the probability of each symbol is given by $P(X = x_k) = p_k$.

- We define the information as

$$I(x_k) = \log \frac{1}{p_k} = -\log(p_k).$$

$0 \le p_k \le 1$

information is non-negative

If logarithm is base 2, information is given in bits.

- Note that $I(x_k) \geq 0$, *i.e.*, non-negative. The equality only holds when $p_k = 1$, which means there is **no uncertainty**, then **no information**.

What is information

- It represents the *surprise* of seeing the outcome (a highly probable outcome is not surprising).

event	probability	surprise
one equals one	1	0 bits
wrong guess on a 4-choice question	3/4	0.415 bits
correct guess on true-false question	1/2	1 bit
correct guess on a 4-choice question	1/4	2 bits
seven on a pair of dice	6/36	2.58 bits
win any prize at Euromilhões	1/24	4.585 bits
win Euromilhões Jackpot	$\approx$ 1/76 million	$\approx$ 26 bits
gamma ray burst mass extinction today	$< 2.7 \cdot 10^{-12}$	$>$ 38 bits

Larger surprise/uncertainty, more information/bits.

Entropy

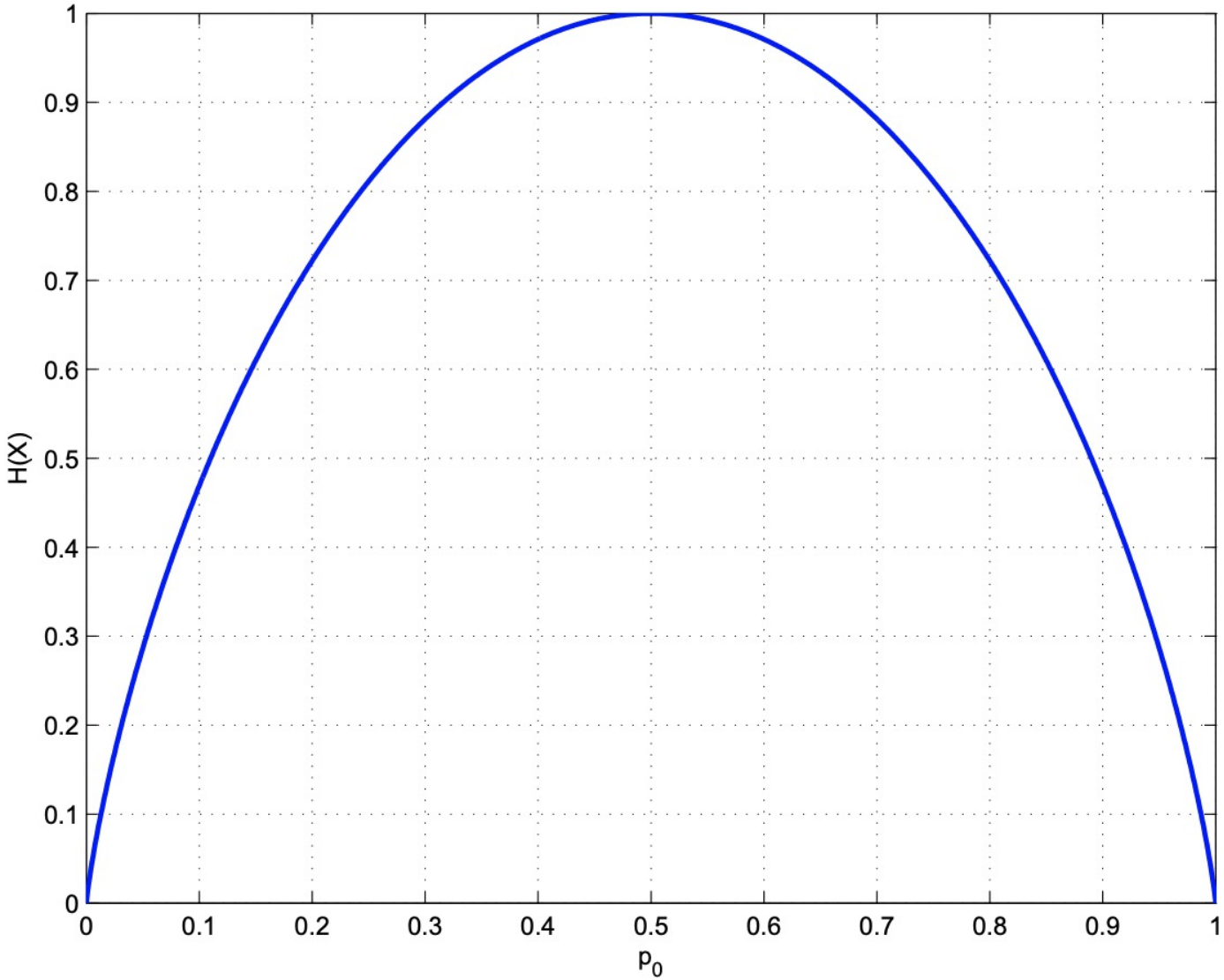
- Entropy (熵) is defined as the **expected value of information** from a source,

$$\begin{aligned} H_P(\mathcal{X}) &= E[I(x_k)] \\ &= \sum_{x_k \in \mathcal{X}} p_k \cdot I(x_k) \\ &= -\sum_{x_k \in \mathcal{X}} p_k \cdot \log(p_k). \end{aligned}$$

- Let $\mathcal{X} = \{0, 1\}$ be a binary source with p_0 and p_1 being the probability of symbols $x_0 = 0$ and $x_1 = 1$, respectively, we have

$$\begin{aligned} H_P(\mathcal{X}) &= E[I(x_k)] \\ &= -p_0 \log p_0 - p_1 \log p_1 \\ &= -p_0 \log p_0 - (1 - p_0) \log (1 - p_0) \end{aligned}$$

Entropy of binary source



Cross Entropy

- Cross-entropy builds upon the idea of entropy. It calculates the number of bits required to represent or transmit an average event from one distribution $P(X)$, compared to another distribution $Q(X)$,

ground tool you know from traing data ← compare → output of machine learning algorithm

$$H_{P,Q}(\mathcal{X}) = -\sum_{x_k \in \mathcal{X}} P(X = x_k) \cdot \log(Q(X = x_k)),$$

  where $P(X = x_k)$ is the probability of the event x_k in $P(X)$, $Q(X = x_k)$ is the probability of event x_k in $Q(X)$.

- Let $\mathcal{X} = \{0, 1\}$ be a binary source with p_0 and p_1 being the probability of symbols $x_0 = 0$ and $x_1 = 1$, respectively, then we have

$$\begin{aligned} H_{P,Q}(\mathcal{X}) &= -p_0 \log q_0 - p_1 \log q_1 \\ &= -p_0 \log q_0 - (1 - p_0) \log (1 - q_0). \end{aligned}$$

Cross Entropy

There are two properties of cross-entropy:

- It is a **non-negative**, *i.e.*, $H_{P,Q}(\mathcal{X}) \geq 0$
- Cross-entropy is not smaller than entropy, *i.e.*, $H_{P,Q}(\mathcal{X}) \geq H_P(\mathcal{X})$, and the equality holds only when $P = Q$.

Exercise: Prove the above two properties
(hint: Jensen inequality and $\log(\cdot)$ is concave)

Reference: https://machinelearningmastery.com/cross-entropy-for-machine-learning/

Relative entropy: Kullback-Leibler divergence

- The relative entropy between two continuous probability density functions $p_X(x)$ and $q_X(x)$ is defined as follows

$$D_{P,Q}(\mathcal{X}) = \int_{x\in\mathcal{X}} p_X(x) \log \frac{p_X(x)}{q_X(x)}. \quad \text{cts.}$$

- It is also called KL divergence, which measures the distance between two distributions.

$\sum p(x) \log_2 \frac{P_X(x)}{q_X(x)}$ discrete

$= \log_2 P_X(x) - \log_2 q_X(x)$ "distance"

Relative entropy: Kullback-Leibler divergence

- Exercise 1: What is a KL divergence for a discrete r.v. ?
- There are two properties of KL divergence:
  - **Non-negativity**: $D_{P,Q}(\mathcal{X}) \geq 0$
  - **Asymmetry**: $D_{P,Q}(\mathcal{X}) \neq D_{Q,P}(\mathcal{X})$
- Exercise 2: Prove the above two properties (hint: Jensen inequal and $\log(\cdot)$ is concave)
- The cross-entropy $H_{P,Q}(\mathcal{X})$ is the entropy of the distribution $H_P(\mathcal{X})$ plus the additional KL divergence $D_{P,Q}(\mathcal{X})$.

$$H_{P,Q}(\mathcal{X}) = H_P(\mathcal{X}) + D_{P,Q}(\mathcal{X})$$

- Exercise 3: Can you prove the equality of the equation?

Reference:

https://machinelearningmastery.com/cross-entropy-for-machine-learning/
https://stats.stackexchange.com/questions/335197/why-kl-divergence-is-non-negative